

Jeremy Welsh

Research Engineer
Differentiable Simulation & Numerical Optimization

📍 San Francisco, CA
✉ jeremy@micromelody.net
☎ (503) 890-1543
in [linkedin.com/in/jeremy-welsh](https://www.linkedin.com/in/jeremy-welsh)
🐙 github.com/jeremyiwk

Summary

Research engineer specializing in computational electromagnetism, PDE-constrained optimization, and fast algorithms for on-instrument hardware control. I work at the level of numerical methods – deriving adjoints, tuning the underlying linear algebra, and implementing it on GPUs – but am motivated by physical results: robust imaging through optimized optics and real-time stabilization of high-voltage instrumentation.

Experience

Software Engineer - Computational Physics | Jun 2025 - Present *Metalenz, Boston, MA (Remote)*

- Accelerated RCWA forward/gradients calculations ~8x double-precision and ~30x mixed-precision, with a ~10x reduction in peak memory by exploiting matrix sparsity, reusing forward-pass matrix factorizations in reverse-mode autodiff, and implementing a mixed-precision eigensolver with CUDA.jl.
- Primary developer of an end-to-end metasurface design platform, integrating design specification, full-wave E&M simulations, gradient-based optimizations, database persistence, and tape-out.
- Robust photonics design: formulated optimality criteria for metasurface-imaging co-optimization to produce designs robust to fabrication errors and sensor noise.
- Polarimetric imaging: built pipelines (demosaicking, super-resolution) converting raw sensor data to polarization images for downstream spoof detection.

Computational Scientist, Advanced Technology | Jun 2022 - Jun 2025 *Thermo Fisher Scientific, Portland, OR*

- Delivered a PFIB alignment algorithm to R&D for integration into an autonomous system for instrument control and health monitoring – an on-tool quasi-Newton method to stabilize PFIB optics, extend aperture lifetime, and reduce the frequency of full-alignment cycles by ~10x.
- Differentiable FIB simulation: built a differentiable charged-particle simulation tool for ion-optics design optimization and aberration measurement – including a 2D boundary element method (BEM) solver for optimizing cross-sectional multipole geometry.
- FIB milling simulation: built a fast 2D level-set model retaining key nonlinearities of FIB milling (surface curvature, beam angle), enabling numerical process optimization.

Graduate Research Assistant | Sep 2020 - Jun 2022 *University of Oregon, Eugene, OR*

- Coarse-grained models for biological macromolecules: used molecular dynamics simulation data to develop a Langevin model of DNA and protein dynamics.

Education

Physics, MS | 2020 - 2022 *University of Oregon, Eugene, OR*

Mathematics and Physics, BS | 2016 - 2020

Skills

Simulation and Numerical Methods: mesh-based and spectral methods for PDEs (RCWA, FDFD/FDTD, FEM, BEM); level-set/Hamilton-Jacobi PDEs; adjoint methods for computational E&M; dense and sparse linear algebra

Optimization: gradient-based methods for constrained/unconstrained, convex/nonconvex problems (MMA, SLSQP, BFGS); closed-loop algorithms for high-voltage hardware control

Languages: Python, Julia, Bash – working knowledge: C++, Rust, Fortran, Mathematica

Libraries/HPC: JAX, PyTorch, Numba, CUDA, CUDA.jl, Metal.jl, Zygote.jl, MPI, Slurm